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Agentic Tools for Water Planning, and How to Catch Them Failing

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The question has changed

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"**Can we build** decision-support tools"? → "**Can we trust** decision-support tools"?

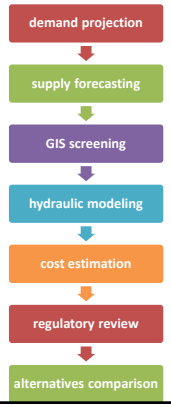
Learning Outcomes:

1. Identify where agentic workflows add value.
2. Recognize failures in data, models, workflow, and assumptions.
3. Select checks that expose failures before decisions are made.

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Water planning is a multi-tool workflow

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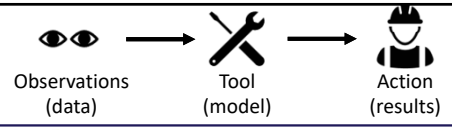


Most planning processes already combine multiple tools and datasets.

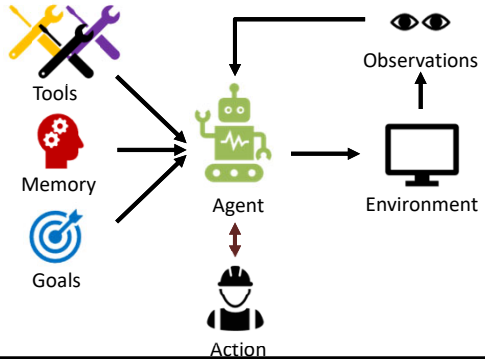
Agentic systems can *orchestrate* existing tools.

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Traditional ML



Agentic AI



"Autonomously" plans which tools to use, in what order, to achieve an outcome

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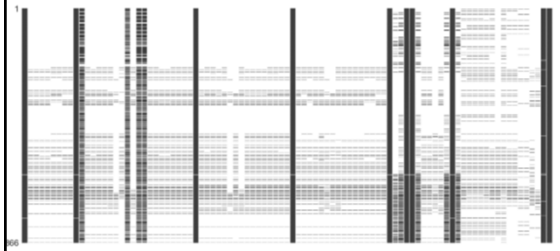
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Where Agentic Tools Fail

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Failure Category #1: Using the Wrong Data

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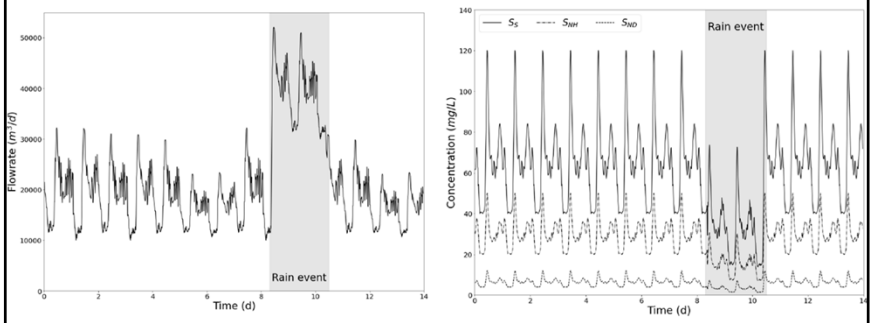
Missingness pattern: black indicates data for a timestep, white is missing.
AlSayed et al., Proceedings of the 10th Water Resource Recovery Modelling Seminar, 2026

- **Coverage:** Are the relevant places, periods, and conditions represented?
- **Compatibility:** Are definitions, units, and spatial scales consistent?
- **Currency:** Can we identify exactly which data version was used and when it was retrieved?

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Failure Category #2: Conditions Outside the Training Domain

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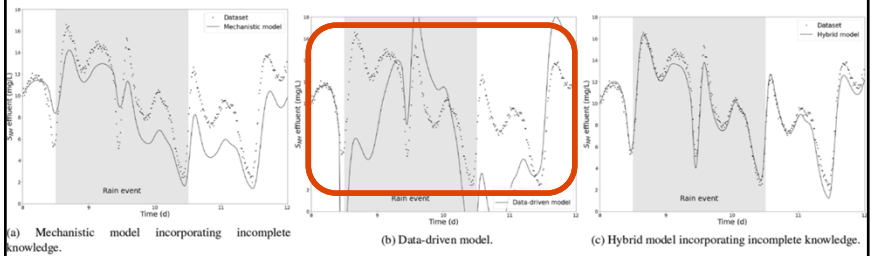


Flowrate and contaminant concentration for **dry weather**, and **wet weather** patterns
Quaghebeur et al., 2023, Water Research

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Failure Category #2: Conditions Outside the Training Domain

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(a) Mechanistic model incorporating incomplete knowledge. (b) Data-driven model. (c) Hybrid model incorporating incomplete knowledge.

Contaminant concentration for **a rain event for a mechanistic, data-driven, and hybrid model**
Quaghebeur et al., 2023, Water Research

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Failure Category #3: Hidden Assumptions

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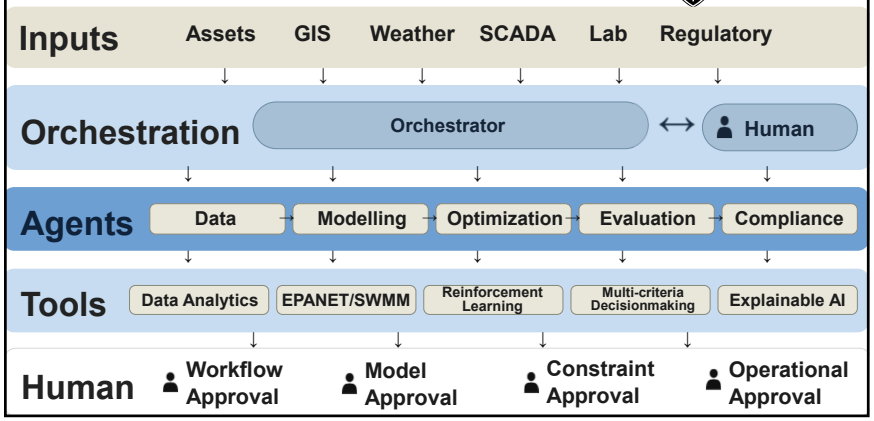
Water planning often fails because *assumptions are inherited* rather than challenged.

A highly automated workflow may simply *propagate an unsupported assumption.*

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Failure Category #4: The Workflow Is Wrong

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Failure Category #4: The Workflow Is Wrong

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- Wrong tool
- Wrong order
- Stale output
- Partial execution

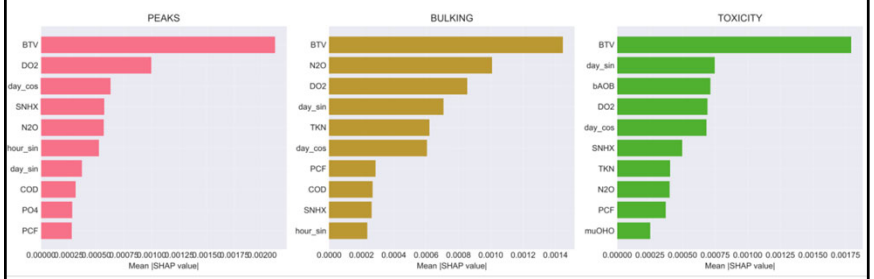
The models may be correct.

The workflow may still be wrong.

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Failure Category #5: Explanations That *Sound* Right

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SHAP plots for agents trained on different events in a wastewater treatment facility
Gomez et al., Proceedings of the 10th Water Resource Recovery Modelling Seminar, 2026

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Failure Category #5: Explanations That *Sound* Right

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Control action	Correlation	Interpretation
DO setpoint, tank 1	0.27	not significant
DO setpoint, tank 2	0.30	not significant
DO setpoint, tank 3	–0.24	GAH!!!!
Waste sludge	0.85	good alignment
Methanol dosing	0.90	good alignment
Reject water	0.84	good alignment
DO actions, mean	**0.11**	**fails for (fast) biological control**
Operational actions, mean	**0.86**	**works for slower processes**

Known control actions and the **correlation with the SHAP value**
Gomez et al., Proceedings of the 10th Water Resource Recovery Modelling Seminar, 2026

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Failure Category #6: One Correct Answer Is Not Validation, Accuracy is Subjective

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What were those metrics calculated on?
- baseline comparisons
- deployment challenges
- optimism bias



Questions planners should ask:
- What is the baseline?
- What conditions were tested?
- Was the tool evaluated after deployment?
- What conditions were excluded?

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A Validation Framework

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What Does Validation Actually Mean?

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Layer	Question
Data	Is the information correct?
Model	Is the analysis correct?
Workflow	Was the process executed correctly?
Decision	Does the output support the stated objective?

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What Does Validation Actually Mean?

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Data • Source and date • Units and schema • Coverage and missingness • Provenance	**Model** • Baseline • Validation domain • Uncertainty • Failure thresholds
Workflow • Tool sequence • Version and configuration • Completion status • Dependency tracking	**Decision** • Objective and constraints • Scenario approval • Unresolved assumptions • Recommendation approval

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Failure Tests

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- Remove data
  - Does the workflow stop or silently impute?
- Change a boundary condition
  - Use a severe drought or high-growth scenario. Does the tool indicate that it has left its validation domain?
- Break a module
  - Force the cost or hydraulic model to fail. Does the workflow withhold the recommendation?

A planning tool should fail visibly and retain enough information to diagnose the failure.

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Demo: an agent plans a recycled water disposal program

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"Our permit is going to prohibit river discharge from May through October. We have 150 acres of pasture. For the wettest summer on record, how much water must we dispose of, how many acres would use it, and how much lagoon storage do we need?"



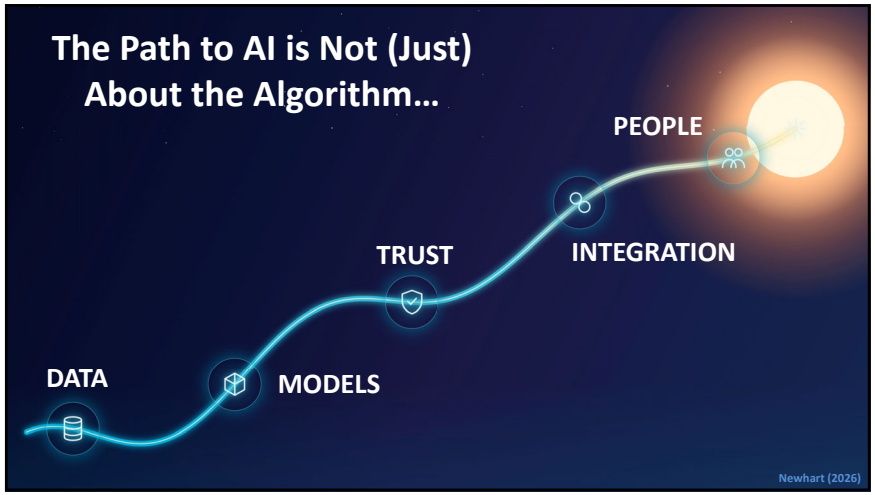
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Before Using an Agentic Planning Tool

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1. What assumptions drive the recommendation?
2. What conditions were represented in the data?
3. What happens outside those conditions?
4. Can the workflow be audited?
5. What human approvals are required?

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Questions?

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